

Canadian Household Credit Risk Monitoring Dashboard

Executive summary | Risk analytics portfolio project

Executive Summary

This project builds a **Canada-level household credit risk monitoring dashboard** using public Canadian macro-financial and insolvency datasets. It helps risk teams monitor household debt stress, identify early-warning indicators, and translate public financial data into executive-level risk insights.

Core business question: Are Canadian households becoming more financially stressed, and which indicators should risk teams monitor?

Project Objective

Create an explainable risk-monitoring dashboard that tracks household credit stress through repayment burden, leverage, insolvency growth, interest pressure, income pressure, and alert-status indicators.

Dashboard Pages

- 1. Executive Risk Overview:** latest risk score, category, trends, and recommended response.
- 2. Credit Stress Drivers:** risk-score components, indicator movement, and data gaps.
- 3. Early Warning Command Center:** alerts, primary driver, status tiles, and response framework.

Risk Score Methodology

The dashboard uses a transparent weighted score from **0 to 100**. It is intended for portfolio-level monitoring, not borrower-level credit decisioning.

Component	Weight	Purpose
Debt service ratio	30%	Measures household repayment burden
Debt-to-income ratio	25%	Measures household leverage
Insolvency growth	20%	Captures realized household financial stress
Interest pressure	15%	Tracks borrowing-cost pressure
Income pressure	10%	Tracks affordability pressure from weak income growth

Risk Categories and Current Alert Status

Score	Category
0-39	Low Risk
40-59	Moderate Risk
60-74	Elevated Risk
75-100	High Risk

Indicator	Status
Debt service ratio	Alert
Debt-to-income	Alert
Insolvency growth	Watch
Interest pressure	Normal
Income pressure	Watch

Key Insight

- Canadian household credit risk moved from Elevated Risk during 2023-2024 to Moderate Risk by late 2025.
- The improvement was mainly driven by slower insolvency growth, easing interest pressure, and a lower debt service ratio compared with recent highs.
- Household vulnerability remains because debt-to-income is still high and repayment pressure remains elevated compared with earlier periods.

Recommended Risk Response

Risk teams should **increase monitoring frequency** and continue reviewing household credit stress indicators, especially household leverage, debt service burden, insolvency momentum, interest-cost pressure, and income growth weakness.

- Monitor debt service ratio and debt-to-income pressure as primary alert areas.
- Track insolvency momentum for renewed deterioration.
- Use Data Gap flags to improve source completeness, refresh logic, and scoring inputs.

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Business Value

This project demonstrates an end-to-end risk analytics workflow that converts public financial data into an explainable monitoring product for stakeholders.

Capability	Demonstrated Output
Data preparation	Cleaned, filtered, aggregated, and joined public household credit and insolvency data
Metric engineering	Created YoY changes, component scores, risk score, and alert statuses
Risk logic	Designed weighted scoring and threshold-based early-warning classifications
Dashboarding	Built executive overview, driver analysis, and command-center alert pages in Power BI
Stakeholder reporting	Provided interpretation, recommended response, and threshold definitions

Early Warning Logic

Status	Meaning	Example Response
Normal	Indicator is within stable range	Continue standard monitoring
Watch	Indicator requires closer monitoring	Increase monitoring frequency
Alert	Indicator shows meaningful stress	Review vulnerable segments
High Alert	Indicator has breached high-risk threshold	Escalate and stress-test portfolios
Data Gap	Required data is unavailable or incomplete	Review source completeness and scoring inputs

Tools Used

Python, Pandas, NumPy, SQL, Power BI, Excel, Statistics Canada data, and Canada Open Government insolvency statistics.

Limitations

This dashboard is designed for **portfolio-level monitoring and educational analysis**. The risk score uses analyst-defined weights and thresholds. In a production financial institution, the scoring logic should be validated against historical delinquency, default, charge-off, borrower-level credit performance, or insolvency outcomes before operational credit decisioning.

Final Takeaway

The dashboard provides a practical, explainable risk-monitoring workflow: current risk status, risk drivers, alert indicators, and recommended response in one Power BI reporting product.

Portfolio Project Outputs: Power BI dashboard (.pbix), dashboard screenshots, processed CSV/XLSX dataset, Python cleaning notebook, SQL scripts, data dictionary, and this executive summary PDF.